

Differential Photometry Plotting Tutorial

Overview

The differential photometry plotting program (`plot_differential_photometry.py`) creates professional-quality light curves from the output of the PhotoCalib pipeline. It produces multi-panel plots showing both differential photometry (target minus reference stars) and raw instrumental magnitudes, allowing you to identify real brightness variations in your target while correcting for atmospheric and instrumental effects.

What is Differential Photometry?

Differential photometry is a technique that measures the brightness of a target star relative to reference stars in the same field. This method automatically corrects for:

- **Atmospheric extinction changes:** Clouds, haze, or airmass variations
- **Instrumental drift:** Temperature changes, focus shifts
- **Sky brightness variations:** Moonlight, twilight effects

By subtracting reference star magnitudes from your target, these common-mode effects cancel out, revealing intrinsic brightness changes in your target like:

- Exoplanet transits
- Variable star pulsations
- Asteroid rotation
- Stellar flares or eclipses

Installation

Ensure you have the required packages:

```
pip install numpy pandas matplotlib
```

Basic Usage

Simple Example: Plotting an Exoplanet Transit

```
python3 plot_differential_photometry.py hatp23_photometry.csv \
-o hatp23_lightcurve.pdf \
--verbose
```

With Custom Settings

```
python3 plot_differential_photometry.py wasp135b_photometry.csv \
```

```
-o wasp135b_transit.pdf \  
--title "WASP-135b Transit on 2025-08-09" \  
--markersize 6 \  
--figsize 14 12 \  
--ylim -0.05 0.05 \  
--verbose
```

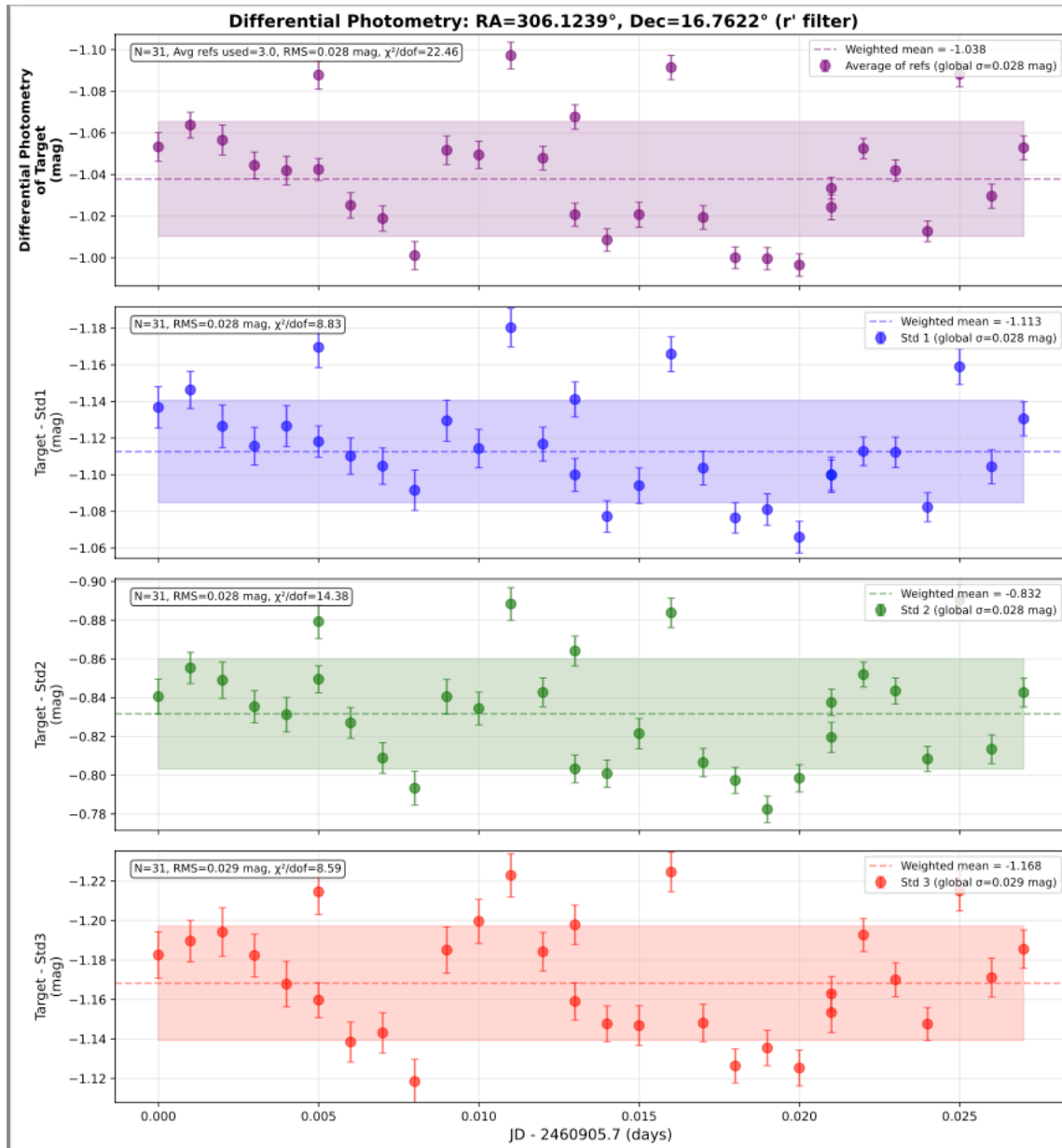
Command Line Arguments

Argument	Default	Description
csv_file	Required	Input CSV from PhotoCalib pipeline
-o, --output	Auto	Output plot filename (PDF/PNG)
--title	Auto	Custom plot title
--show	False	Display plot interactively
--markersize	8	Size of data points
--figsize	12 10	Figure size in inches (width height)
--ylim	Auto	Y-axis limits for all panels (min max)
--verbose	False	Print detailed statistics

Understanding the Output Plots

The program creates a two-page PDF with complementary views of your data:

Page 1: Differential Photometry (4 panels)



This page shows the differential photometry - the heart of precision photometry:

Top Panel: Average of All References

- **Purple points:** Target magnitude minus the weighted average of all valid reference stars
- **Dashed line:** Weighted mean of the differential photometry
- **Shaded region:** $\pm 1\sigma$ uncertainty band
- **Statistics box:**
 - N: Number of valid measurements

- Avg refs used: Average number of reference stars per point
- RMS: Root-mean-square scatter
- χ^2/dof : Reduced chi-squared (should be ~ 1 for good error estimates)

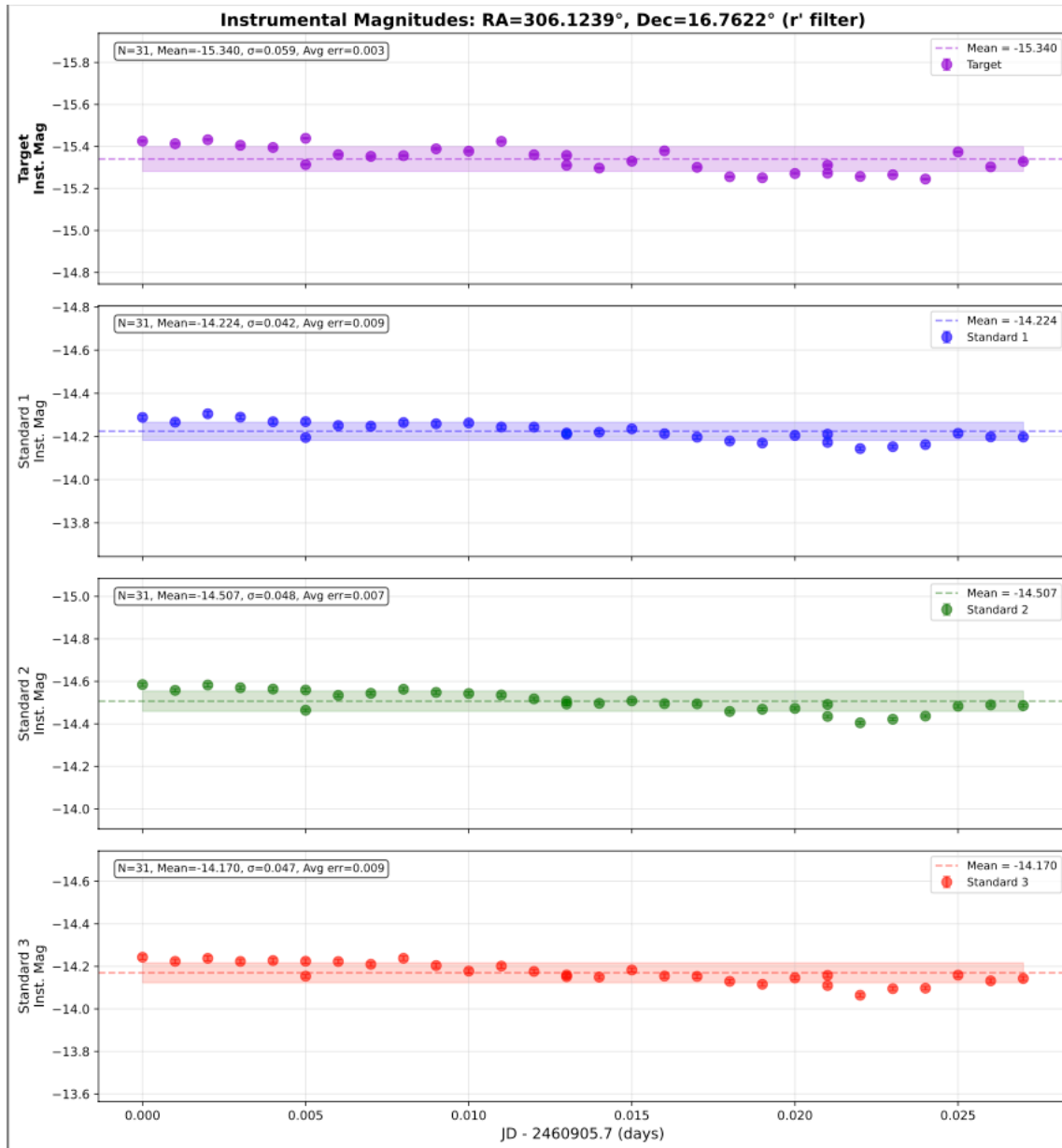
Panels 2-4: Individual Reference Stars

- **Blue/Green/Red points:** Target minus each reference star (Std1, Std2, Std3)
- **Error bars:** Individual photometric uncertainties propagated from both stars
- **Global σ :** Standard deviation of the entire dataset
- **Weighted mean:** Average weighted by individual uncertainties

What to look for:

- **Consistent patterns across panels:** Real variations in your target
- **Variations in only one panel:** That reference star may be variable
- **Increasing scatter over time:** Deteriorating conditions or tracking issues
- $\chi^2/\text{dof} \gg 1$: Underestimated errors or unmodeled variability
- $\chi^2/\text{dof} \ll 1$: Overestimated errors

Page 2: Instrumental Magnitudes (4 panels)



This page shows the raw instrumental magnitudes before differential correction:

Panel 1: Target Star

- Shows the uncorrected brightness of your target
- Large variations often due to atmospheric changes
- Look for saturation warnings or sudden jumps

Panels 2-4: Reference Stars

- Should show similar patterns to the target (atmospheric effects)
- Different patterns indicate that star may be variable
- Check that reference stars track each other well

What to look for:

- **Parallel trends:** All stars affected equally by atmosphere
- **Divergent behavior:** Possible variable star or measurement problem
- **Sudden jumps:** Clouds, focus changes, or tracking errors
- **Gradual slopes:** Atmospheric extinction changes with airmass

Interpreting Your Results

Example 1: Clean Exoplanet Transit

In this example from HAT-P-23b, we see:

DIFFERENTIAL PHOTOMETRY (Page 1):

- Top panel: Clear transit ingress starting at JD 0.09
- Transit depth: ~0.3 magnitudes (3% flux drop)
- All three reference stars show the same pattern
- Low RMS (~0.078-0.120 mag) indicates good precision
- χ^2/dof values of 13-20 suggest some unmodeled variability

INSTRUMENTAL MAGNITUDES (Page 2):

- Target shows transit plus atmospheric trends
- Reference stars show only atmospheric trends
- All stars brighten together early (improving conditions)
- Reference stars remain stable during target's transit

Statistical Interpretation

The program provides several key statistics:

1. **RMS (Root Mean Square)**
 - Typical good values: 0.001-0.010 mag for bright stars
 - 0.010-0.050 mag for moderate brightness
 - 0.100 mag indicates poor conditions or faint stars
2. **χ^2/dof (Reduced Chi-Squared)**
 - ~1.0: Error bars correctly estimated
 - 1.0: Underestimated errors or intrinsic variability
 - <<1.0: Overestimated errors
3. **Global σ vs Individual Errors**
 - Similar values: Errors well-estimated
 - Global $\sigma \gg$ individual errors: Additional noise sources

Practical Examples

Detecting an Exoplanet Transit

```
# Process photometry
python3 photocalib_robust_fixed.py ~/data/WASP-135b -f "r'" \
    -o wasp135b_phot.csv \
    --target-ra "17:49:08.39" --target-dec "+29:52:44.8"

# Create light curve
python3 plot_differential_photometry.py wasp135b_phot.csv \
    -o wasp135b_transit.pdf \
    --title "WASP-135b Transit - 2025-08-09" \
    --ylim -0.02 0.01 \
    --verbose
```

Expected output:

- Flat baseline before/after transit
- Smooth ingress/egress lasting ~20 minutes
- Flat bottom during transit
- Consistent depth across all reference stars

Monitoring a Variable Star

```
python3 plot_differential_photometry.py rr_lyrae_phot.csv \
    -o rr_lyrae_pulsation.pdf \
    --title "RR Lyrae Pulsation Cycle" \
    --markersize 4 \
    --verbose
```

Expected output:

- Periodic variations with consistent amplitude
- Same pattern in all differential panels
- Period typically 0.2-1.0 days for RR Lyrae stars

Asteroid Light Curve

```
python3 plot_differential_photometry.py asteroid_433_phot.csv \
    -o eros_rotation.pdf \
    --title "433 Eros Rotation - 5.27 hour period" \
    --figsize 16 10 \
    --verbose
```

Expected output:

- Double-peaked light curve (elongated body)
- Period of several hours
- Amplitude 0.1-1.5 magnitudes

Advanced Analysis

Examining Output Statistics

With `--verbose`, the program prints detailed statistics:

```
DIFFERENTIAL PHOTOMETRY SUMMARY
```

```
=====
Standard 1:
  Points with valid data: 97
  Mean differential mag: -0.689
  Global std dev: 0.097
  Mean individual error: 0.032
  Reduced  $\chi^2$ : 14.29
  RMS: 0.120
```

```
Standard 2:
  Points with valid data: 97
  Mean differential mag: -0.602
  Global std dev: 0.192
  Mean individual error: 0.029
  Reduced  $\chi^2$ : 19.47
  RMS: 0.204
```

This tells you:

- **Standard 1** is the most stable reference (lowest RMS)
- **Standard 2** has higher scatter (may be slightly variable)
- High χ^2 values suggest additional noise sources

Identifying Problems

All panels show different patterns

- Reference stars may be variable
- Check diagnostic PDFs from PhotoCalib
- Consider re-running with different reference stars

Scatter increases over time

```
# Plot with wider Y limits to see full extent
python3 plot_differential_photometry.py data.csv \
  --ylim -0.5 0.5 \
  --verbose
```

- Likely due to weather or tracking issues
- Check FWHM values in CSV for focus problems

Missing data points

- Some reference stars not found in all images
- Check 'n_standards' column in CSV
- May need to adjust search radius in PhotoCalib

Quality Assessment Checklist

✓ Good differential photometry shows:

- Consistent patterns across all reference stars
- RMS < 0.01 mag for bright targets
- χ^2/dof between 0.5 and 2.0
- Stable baseline regions
- Smooth variations (not noisy)

✗ Problems indicated by:

- Different patterns in each panel
- RMS > 0.1 mag for bright stars
- χ^2/dof > 10 or < 0.1
- Sudden jumps or gaps
- Increasing scatter over time

Output File Formats

The program supports multiple output formats:

```
# PDF (vector graphics, best for publications)
python3 plot_differential_photometry.py data.csv -o lightcurve.pdf

# PNG (raster graphics, good for presentations)
python3 plot_differential_photometry.py data.csv -o lightcurve.png

# Interactive display (for exploration)
python3 plot_differential_photometry.py data.csv --show
```

Tips for Best Results

1. **Optimize Y-axis scaling:** Use `--ylim` to zoom in on small variations
2. `--ylim -0.01 0.01` # For detecting 1% transits
3. **Adjust marker size** for dense or sparse data:
4. `--markersize 3` # For hundreds of points
5. `--markersize 12` # For sparse data
6. **Create publication-quality figures:**
7. `--figsize 8 10` # Single column in journal
8. `--figsize 16 10` # Full page width
9. **Compare multiple nights:**
10. `for night in night1.csv night2.csv night3.csv; do`

```
11. python3 plot_differential_photometry.py $night \  
12. -o ${night%.csv}_diff.pdf --verbose \  
13. done
```

Troubleshooting

"No valid data for Standard X"

- That reference star wasn't found in enough images
- Check PhotoCalib diagnostic PDFs
- May need to select different reference stars

Error bars too large/small

- Check SNR values in CSV
- Verify aperture size was appropriate
- Consider weather conditions

Unexpected trends in differential photometry

- One reference star may be variable
- Check instrumental magnitude plot (page 2)
- Compare behavior of each reference star

Conclusion

The differential photometry plotter transforms raw photometry into science-ready light curves. By showing both differential and instrumental magnitudes, it helps you distinguish real astrophysical variations from systematic effects. The statistical measures and error propagation ensure you can assess the significance of any detected variations, whether hunting for exoplanet transits, tracking variable stars, or measuring asteroid rotations.